

2023

- ① Explain the in detail why is not good statistical procedure to perform several t-test on pairs of mean. When several means are to be compared suggest the alternative procedure.

performing several t-test on pairs of means increase.

- **Type I Error Rate:-**

The probability of rejecting a true null hypothesis increases with multiple tests.

- **Inflated Error Rate:-**

As the number of tests increase, the likelihood of at least one false positive result also increase.

Alternative procedure
→ **Analysis of Variance (ANOVA)**

ANOVA is a suitable alternative for comparing multiple means. It:

- **Control Type I Error Rate:-**

ANOVA maintains a fixed significance level (α) for the entire analysis.

- **Compares Multiple Means:-**

ANOVA assesses whether there are significant differences between three or more group means.

② In a linear regression problem the following results are obtained

$$Y = 12.13 - 0.1608x$$

$$\sum (Y - \hat{Y})^2 = 1160, \quad \sum (x - \bar{x})^2 = 1300$$

$$n = 24$$

Find 95% Confidence interval for α and β .

Solution:-

The 95% Confidence

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interval the β is given

$$b \pm t_{\frac{\alpha}{2}(n-2)} S_b$$

$$b = -0.1608$$

$$t_{\frac{\alpha}{2}(n-2)} = t_{\frac{0.05}{2}(24-2)}$$

$$= t_{0.025}(22)$$

$$= 2.0711$$

$$S_b^2 = \frac{\sum x_i^2 - \frac{(\sum x_i)^2}{n}}{\sum (x_i - \bar{x})^2} \cdot \frac{\sum (y_i - \hat{y}_i)^2}{(n-2) \sum (x_i - \bar{x})^2}$$

put the values -

$$= \frac{1160}{(24-2)(1300)}$$

$$= \frac{1160}{(22)(1300)}$$

$$= \frac{1160}{28600} = 0.040$$

$$\sqrt{S_b^2} = \sqrt{0.040}$$

$$S_b = 0.20$$

$$-0.1608 \pm (2.074)(0.20)$$

$$-0.1608 \pm 0.4148$$

$$-0.1608 + 0.4148, -0.1608 - 0.4148$$

$$0.254, -0.5756$$

$$(0.254, -0.5756)$$

Hence the desired 95% Confidence interval for population coefficient β is $(0.254, -0.5756)$.

The 95% confidence interval for α is given by

$$\alpha \pm t_{0.05}(n-2) \frac{S_a}{2}$$

$$\alpha = 12.13 \quad \text{PUACP}$$

$$t_{\frac{0.05}{2}(n-2)} = t_{0.025}(24-2)$$

$$t_{0.025}(22)$$

$$= 2.074$$

$$S_a = S_{y \cdot x} \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{\sum (x - \bar{x})^2}}$$

$$= \frac{\sum (y - \hat{y})^2}{n-2} \sqrt{\frac{1}{24} + \frac{\bar{x}^2}{\sum (x - \bar{x})^2}}$$

put the values

$$S_a = \frac{1160}{22} \sqrt{\frac{1}{24} \frac{\bar{x}^2}{1300}}$$

$$S_a = 52.73 \sqrt{\frac{0.042 + \bar{x}^2}{1300}}$$

Since the value of $\bar{x}$ is not given. So we cannot calculate the value of S_a .

③ What is Coefficient of determination is multiple regression. What it explain.

See notes Q 18

④ Test whether there is any association between the results of preliminary and those of the final test

Preliminary Test		
Categories	pass	Fail
Final pass	605	135
Final Fail	195	65

Solution:-

Hypothesis:-

H_0 : The results of the final test are independent of those of the preliminary test.

H_1 : The results of the final test are not independent of those of the preliminary test.

Level of significance:-

$$\alpha = 0.05$$

Test-statistics:-

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Calculation:-

Categories	preliminary Test		Total
	pass	Fail	
Final pass	605	135	740
Final Fail	195	65	260
Total	800	200	1000

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Then the calculate χ^2

observed frequency O_{ij}	expected frequency E_{ij}	$(O_{ij} - E_{ij})$	$(O_{ij} - E_{ij})^2$	$\frac{(O_{ij} - E_{ij})^2}{E_{ij}}$
605	$\frac{800 \times 740}{1000} = 592$	13	169	0.285
195	$\frac{800 \times 260}{1000} = 208$	-13	169	0.8125
135	$\frac{740 \times 200}{1000} = 148$	-13	169	1.1418
65	$\frac{200 \times 260}{1000} = 52$	13	169	3.25
Total 1000	1000	---	---	= 5.4893

$$\chi^2 = 5.4893$$

Critical region:-

$$\chi^2_{\text{PUAP}} \geq \chi^2_{0.05} (r-1)(c-1)$$

$$\chi^2 \geq \chi^2_{0.05} (2-1)(2-1)$$

$$\chi^2 \geq \chi^2_{0.05} (1)(1)$$

$$\chi^2 \geq \chi^2_{0.05} (1)$$

$$\chi^2 \geq 3.84$$

$$5.4893 \geq 3.84$$

Conclusion:-

Since the calculated value of Chi-Square (χ^2) is greater than the critical region

So, the null hypothesis (H_0) is reject and alternative hypothesis (H_1) is accepted.

- ⑤ Differentiate between partial and multiple Correlation.

See the notes.

- ⑥ Find the Simple Correlation Coefficient r_{12} , r_{13} and r_{23} for the following information

$$b_{12} = -0.1 \quad b_{21} = -0.4 \quad b_{13} = 0.27 \\ b_{31} = 0.6 \quad b_{23} = 0.67 \quad b_{32} = 0.38$$

Solution:-

Simple Correlation Coefficient

$$r_{12} = -\sqrt{r_{12} \times r_{21}}$$

$$r_{13} = \sqrt{r_{13} \times r_{31}}$$

$$r_{23} = \sqrt{r_{23} \times r_{32}}$$

$$r_{12} = -\sqrt{r_{12} \times r_{21}}$$

$$= -\sqrt{(-0.1) \times (-0.4)}$$

$$= -\sqrt{0.04}$$

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$$r_{12} = -0.2$$

$$\begin{aligned} r_{13} &= \sqrt{b_{13} \times b_{31}} \\ &= \sqrt{0.27 \times 0.6} \\ &= \sqrt{0.162} \end{aligned}$$

$$r_{13} = 0.402$$

$$\begin{aligned} r_{23} &= \sqrt{r_{23} \times r_{32}} \\ &= \sqrt{0.67 \times 0.38} \\ &= \sqrt{0.2546} \end{aligned}$$

$$r_{23} = 0.5046$$

Long Questions

- ① Following in the plan of a field layout of the testing four varieties A, B, C and D of wheat in each 5 blocks. The plot yields in pound are also indicated there is

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Blocks				
1	2	3	4	5
D 29.3	B 33.0	D 29.8	B 36.8	D 28.8
B 33.3	A 34.0	A 34.3	A 35.0	C 35.8
C 30.8	C 34.3	B 36.3	D 28.0	B 34.5
A 32.3	D 26.0	C 35.3	C 32.3	A 36.5

Solution:-**Hypothesis:-**

H_0 : All the four varieties are equal

H_1 : All the four varieties are not equal.

Significance level:-

$$\alpha = 0.05$$

Test- statistics:-

$$F_1 = \frac{S_1^2}{S_3^2}$$

$$F_2 = \frac{S_2^2}{S_3^2}$$

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Calculation:-

Blocks	Treatments				T_i	T_i^2	$\sum x_{ij}^2$
	A	B	C	D			
1	32.3 (1043.29)	33.3 (1108.89)	30.8 (948.64)	29.3 (858.49)	125.7 127.3	15800.49	3934.31
2	34.0 (1156)	33.0 (1089)	34.3 (1176.49)	26.0 (676)	127.3	16205.29	4097.49
3	34.3 (1176.49)	36.3 (1317.69)	35.3 (1246.09)	29.8 (888.04)	135.7	18414.49	4628.31
4	35.0 1225	36.8 (1354.24)	32.3 (1043.29)	28.0 (784)	132.1	17450.41	4406.53
5	36.5 1332.25	34.5 (1190.25)	35.8 (1281.64)	28.8 (829.44)	135.6	18387.36	4633.58
T_j	172.1	173.9	168.5	141.9	656.4	86258.04	2172
T_j^2	29618.41	30246.21	28392.25	20135.61	108387.49		↓
$\sum x_{ij}$	5933.03	6060.07	5696.15	4035.97	21725.22	→	checked

$$C.F = \frac{T^2}{rc} = \frac{(656.4)^2}{20}$$

$$= \frac{430860.96}{20}$$

$$C.F = 21543.048$$

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$$\begin{aligned}\text{Total ss} &= \sum x_{ij}^2 - C \cdot F \\ &= 21725.22 - 21543.048\end{aligned}$$

$$\boxed{\text{Total ss} = 182.172}$$

$$\begin{aligned}\text{Treatment ss} &= \frac{T_j^2}{r} - C \cdot F \\ &= \frac{102387.49}{5} - 21543.048\end{aligned}$$

$$\begin{aligned}&= 20477.498 - 21543.048 \\ \text{Treatment ss} &= 134.45\end{aligned}$$

$$\begin{aligned}\text{Block ss} &= \frac{\sum l_i^2}{k} - C \cdot F \\ &= \frac{86258.84}{4} - 21543.048\end{aligned}$$

$$\begin{aligned}&= 21564.51 - 21543.048 \\ \text{Block ss} &= 21.462\end{aligned}$$

$$\begin{aligned}\text{Error ss} &= \text{Total ss} - (\text{Treatment ss} + \text{Block ss}) \\ &= 182.172 - (134.45 + 21.462) \\ &= 182.172 - 155.912\end{aligned}$$

$$\boxed{\text{Error ss} = 26.26}$$

Solution:-

⑨ The estimated multiple linear regression equation is

$$\hat{X}_1 = a + b_1 X_2 + b_2 X_3$$

Three normal equations are

$$\sum X_1 = na + b_1 \sum X_2 + b_2 \sum X_3$$

$$\sum X_1 X_2 = a \sum X_2 + b_1 \sum X_2^2 + b_2 \sum X_2 X_3$$

$$\sum X_1 X_3 = a \sum X_3 + b_1 \sum X_2 X_3 + b_2 \sum X_3^2$$

X_1	X_2	X_3	X_2^2	X_3^2	$X_2 X_3$	$X_1 X_2$	$X_1 X_3$	X_1^2
1	1	2	1	4	2	1	2	1
4	8	8	64	64	54	32	32	16
1	3	1	9	1	3	1	1	1
3	5	7	25	49	35	15	21	9
2	6	4	36	16	24	12	8	4
4	10	6	100	36	60	40	24	16
Total	15	33	28	235	170	103	88	47

All the calculated values put in normal equation

$$6a + 33b_1 + 28b_2 = 15 \quad \text{--- (1)}$$

$$33a + 235b_1 + 188b_2 = 103 \quad \text{--- (2)}$$

$$28a + 188b_1 + 170b_2 = 88 \quad \text{--- (3)}$$

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Multiply by eq ① by -11
and eq ② by 2

$$66a + 363b_1 + 308b_2 = 165 \quad \text{--- (4)}$$

$$66a + 470b_1 + 376b_2 = 206 \quad \text{--- (5)}$$

Subtract from eq (5) to eq (4)

$$\begin{array}{r} 66a + 470b_1 + 376b_2 = 206 \\ - 66a + 363b_1 + 308b_2 = 165 \\ \hline 107b_1 + 68b_2 = 41 \end{array}$$

$$107b_1 + 68b_2 = 41$$

$$107b_1 = 41 - 68b_2$$

$$b_1 = \frac{41}{107} - \frac{68}{107}b_2$$

$$\boxed{b_1 = 0.383 - 0.6355b_2}$$

The value of b_1 put in
eq (3)

$$\begin{aligned} 28a + 188(0.383 - 0.6355b_2) + 170b_2 &= 88 \\ &= 88 \end{aligned}$$

$$28a + 72.004 - 119.474b_2 + 170b_2 = 88$$

$$28a + 50.526b_2 = 88 - 72.004$$

$$28a + 50.526b_2 = 15.996$$

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ANOVA Table

Sources of variations	d.f	SS	MS	Compute F
TSS	$(Y-1)$ 3	134.45	44.82	$F_1 = 20.47$
BlockSS	$(K-1)$ 4	21.462	5.37	$F_2 = 2.45$
ErrorSS	12	26.26	2.19	---
Total	19	182.172	---	---

Critical regions:-

$$F_1 \geq F_{0.05}(3, 12)$$

$$F_1 \geq 3.49$$

$$20.47 \geq 3.49$$

$$F_2 \geq F_{0.05}(4, 12)$$

$$F_2 \geq 3.26$$

$$2.45 \geq 3.26$$

Conclusion:-

Since the Calculated

value of F_1 is greater than the critical region so the null hypothesis of F_1 is reject.

The value of calculated F_2 is less than the critical region so, the null hypothesis is accept.

② Given the following data

x_1	1	4	1	3	2	4
x_2	1	8	3	5	6	10
x_3	2	8	4	4	4	6

① Find the least square regression line where x_1 dependent variable and x_2 and x_3 are independent variable.

② Calculate the Coefficient of multiple determination and multiple correlation and interpret the result.

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$$28a = 15.996 - 50.526b_2$$

$$a = \frac{15.996}{28} - \frac{50.526}{28}b_2$$

$$a = 0.5712 - 1.8045b_2$$

The value of a and b_1 put
in eq (4)

$$66(0.5712 - 1.8045b_2) + 363(0.383 - 0.6355b_2) + 308b_2 = 165$$

$$37.699 - 119.097b_2 + 139.629 - 230.69b_2 + 308b_2 = 165$$

$$176.728 - 41.787b_2 = 165$$

$$-41.787b_2 = 165 - 176.728$$

$$-41.787b_2 = -11.728$$

$$b_2 = \frac{11.728}{41.787}$$

$$b_2 = 0.280$$

put the value of b_2
in b_1 and a

$$b_1 = 0.393 - 0.6355(0.280) \\ = 0.393 - 0.17794$$

$$\boxed{b_1 = 0.205}$$

$$a = 0.5712 - 1.2045(0.280)$$

$$a = 0.5712 - 0.50526$$

$$\boxed{a = 0.065}$$

All the values put in regression equation

$$X_1 = 0.065 + 0.205X_2 + 0.280X_3$$

(b) The Co-efficient of multiple determination $R^2_{1.23}$ is Computed as

$$R^2_{1.23} = \frac{\sum (\hat{X}_1 - \bar{X}_1)^2}{\sum (X_1 - \bar{X}_1)^2}$$

$$R^2_{1.23} = \frac{a \sum X_1 + b_1 \sum X_1 X_2 + b_2 \sum X_1 X_3 - (\sum X_1)^2/n}{\sum X_1^2 - (\sum X_1)^2/n}$$

put the values in formula

$$R_{1.22}^2 = \frac{(0.06)(15) + (0.205)(103) + (0.280)(88) - (15)^2}{47 - (15)^2 / 6}$$

$$= \frac{0.9 + 21.115 + 24.64 - 225 / 6}{47 - 225 / 6}$$

$$= \frac{0.9 + 21.115 + 24.64 - 37.5}{47 - 37.5}$$

$$= \frac{9.155}{9.5}$$

$$R_{1.23}^2 = 0.9636$$

$$\sqrt{R_{1.23}^2} = \sqrt{0.9636}$$

$$R_{1.23} = 0.9816$$

- ③ Records taken of the number of male and female births in 800 families having four children are as

No. of male births	0	1	2	3	4
Families	32	178	290	236	64

Test whether the data are consistent with the hypothesis

that the binomial law holds and that the chance of a male birth is equal to that of a female birth, that $p=q=0.5$

Solution:-

Hypothesis:-

H_0 : The population distribution is a binomial with $n=4$ and $p=0.5$

H_1 : The population distribution is not a binomial with $p=0.5$

Level of Significance:-

$$\alpha = 0.05$$

Test-statistics:-

$$\chi^2 = \sum \frac{(o_i - e_i)^2}{e_i}$$

Calculation:-

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No. of male births	Families (o_i)	expected frequency e_i	($o - e_i$)	($o - e_i$) ²	($o - e$) ² e_i
0	32	50	-18	324	6.48
1	178	200	-22	484	2.42
2	290	300	-10	100	0.33
3	236	200	36	1296	6.48
4	64	50	14	196	3.92
Total	800	800			$\chi^2 = 19.63$

Find expected frequency

According to binomial
Distribution.

$${}^n C_x p^x q^{n-x}$$

$$x=0$$

$$\begin{aligned}
 &= {}^4 C_0 (0.5)^0 (0.5)^{4-0} \times 800 \\
 &= (1)(1)(0.0625) \times 800 \\
 &= 50
 \end{aligned}$$

$$x=1$$

$$\begin{aligned}
 &= {}^4 C_1 (0.5)^1 (0.5)^{4-1} \times 800 \\
 &= (4)(0.5)(0.125) \times 800 \\
 &= 0.25 \times 800 \\
 &= 200
 \end{aligned}$$

$$n=2$$

$$\begin{aligned}
 &= {}^4C_2 (0.5)^2 (0.5)^{4-2} \times 800 \\
 &= (6)(0.25)(0.25) \times 800 \\
 &= 0.375 \times 800 \\
 &= 300
 \end{aligned}$$

$$n=3$$

$$\begin{aligned}
 &= {}^4C_3 (0.5)^3 (0.5)^{4-3} \times 800 \\
 &= (4)(0.125)(0.5) \times 800 \\
 &= 0.25 \times 800 \\
 &= 200
 \end{aligned}$$

$$n=4$$

$$\begin{aligned}
 &= {}^4C_4 (0.5)^4 (0.5)^{4-4} \times 800 \\
 &= (1)(0.0625)(1) \times 800 \\
 &= 50
 \end{aligned}$$

Critical region:-

$$\chi^2 \geq \chi_{0.05}(k-1)$$

$$\chi^2 \geq \chi_{0.05}(5-1)$$

$$\chi^2 \geq \chi_{0.05}(4)$$

$$\chi^2 \geq 9.49$$

$$19.63 \geq 9.49$$

Conclusion:-

Since the
calculated value of χ^2

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is greater than the critical region. So, the null hypothesis of binomial distribution is rejected and Accept the alternative hypothesis.



2024

- ① What are basic principle of experimental design

Answer see notes

- ② Fit Multiple regression
 $Y = a + bx_1 + cx_2$ To
 following data

$$\sum y = 89, \sum x_1 = 30, \sum x_2 = 52$$

$$\sum x_1^2 = 283, \sum x_2^2 = 582$$

$$\sum x_1 x_2 = 351$$

$$\sum x_1 y = 619, \sum x_2 y = 1007$$

$$n = 10$$

Solution:-

Multiple regression
Equation:-

$$y = a + bx_1 + cx_2$$

Normal Equations-

$$\bullet \sum y = na + b \sum x_1 + c \sum x_2 \quad \text{--- (1)}$$

$$\bullet \sum x_1 y = a \sum x_1 + b \sum x_1^2 + c \sum x_1 x_2 \quad \text{--- (2)}$$

$$\bullet \sum x_2 y = a \sum x_2 + b \sum x_1 x_2 + c \sum x_2^2 \quad \text{--- (3)}$$

⇒ put the values in equation

①

$$89 = 10a + 30b + 52c$$

⇒ put the values in eq ②

$$619 = 30a + 283b + 351c$$

⇒ put the values in eq ③

$$1007 = 52a + 351b + 582c$$

$$89 = 10a + 30b + 52c \quad \text{--- ④}$$

$$619 = 30a + 283b + 351c \quad \text{--- ⑤}$$

$$1007 = 52a + 351b + 582c \quad \text{--- ⑥}$$

Equation - 4

$$89 = 10a + 30b + 52c$$

$$10a = 89 - 30b - 52c$$

divide by 10 on both sides

$$\frac{10a}{10} = \frac{89}{10} - \frac{30b}{10} - \frac{52c}{10}$$

$$a = 8.9 - 3b - 5.2c$$

So, put the value^a of eq ⑤ and ⑥

$$619 = 30(8.9 - 3b - 5.2c) + 283b + 351c$$

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$$619 = 267 - 90b - 156c + 283b + 351c$$

$$619 = 267 + 193b + 195c$$

$$619 - 267 = 193b + 195c$$

$$352 = 193b + 195c \quad \text{--- (7)}$$

$$352 - 195c = 193b$$

$$b = \frac{352}{193} - \frac{195}{193}c$$

$$b = 1.823 - 1.0103c$$

put the value of b in eq (6)

$$1007 = 52(8.9 - 3b - 5.2c) + 351b + 582c$$

$$1007 = 462.8 - 156b - 270.4c + 351b + 582c$$

$$1007 - 462.8 = 195b + 311.6c$$

$$544.2 = 195b + 311.6c \quad \text{--- (8)}$$

subtract eq (7) from eq (8)

$$352 = 193b + 195c$$

$$-544.2 = 195b + 311.6c$$

$$-192.2 = -2b - 116.6c$$

$$-(192.2) = -(2b + 116.6c)$$

$$192.2 = 2b + 116.6c$$

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put the value of b

$$192.2 = 2(1.823 - 1.0103C) + 116.6C$$

$$192.2 = 3.646 - 2.0206C + 116.6C$$

$$192.2 - 3.646 = -2.0206C + 116.6C$$

$$188.554 = 114.5794C$$

$$188.554 = C$$

$$114.5794$$

$$C = 1.65$$

So,

$$b = 1.823 - 1.0103C$$

put the value of C

$$b = 1.823 - 1.0103(1.65)$$

$$b = 1.823 - 1.666$$

$$b = 0.157$$

put the value of b and C
in a

$$a = 8.9 - 3b - 5.2C$$

$$a = 8.9 - 3(0.157) - 5.2(1.65)$$

$$a = 8.9 - 0.471 - 8.58$$

$$a = -0.151$$

The value of a, b, C put in
Multiple regression equation

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$y = -0.151 + 0.157x_1 + 1.65x_2$

③ write down the layout of Latin Square design with 5 Treatments.

Latin Square Design is an experimental design used when there are two blocking variables, such as row and columns and each treatment must appear once in each row and column.

For in 5 treatments (say A, B, C, D, E) draw a 5×5 grid where

- Each row represents a level of one blocking factor

- Each column represents another

- Each treatment appear exactly one in each row and each column.

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	Column				
	1	2	3	4	5
Row 1	A	B	C	D	E
Row 2	B	C	D	E	A
3	C	D	E	A	B
4	D	E	A	B	C
5	E	A	B	C	D

④ (Three varieties of Latin Square design with) x

④ Three Varieties A, B, C of a Crop are tested in CRD (Complete randomized design) experimental yield and state your conclusion.

A	B	C
813	647	713
C	B	A
814	759	795
A	C	B
795	652	598

Solution:-

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Data

Variety	Yield
A	813
B	647
C	713
C	814
B	759
A	795
A	795
C	652
B	598

Calculation:-

calculate
the mean yield for each
variety

A:-

$$A = \frac{813 + 795 + 795}{3}$$

$$A = \frac{2403}{3}$$

$$A = 801$$

B:-

$$B = \frac{647 + 759 + 598}{3}$$

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- ⑤ Differentiate between Simple and partial Correlation.

See notes

- ⑥ Find the least value of r in a sample of 18 pairs from a bivariate normal population that is significant at 5% level.

Solution:-

The test statistics for the significance of r is

$$t = \frac{r \sqrt{n-2}}{\sqrt{1-r^2}}$$

$$n = 18$$

$$t = \frac{r \sqrt{18-2}}{\sqrt{1-r^2}}$$

$$t = \frac{r \sqrt{16}}{\sqrt{1-r^2}}$$

$$t = \frac{r \sqrt{4^2}}{\sqrt{1-r^2}}$$

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$$t = \frac{4x}{\sqrt{1-x^2}}$$

$$\sqrt{1-x^2}$$

For a significant value of r at 0.05 level t , should be

$$t > \frac{t_{0.05}(n-2)}{2}$$

$$t > t_{0.025}(18-2)$$

$$t > t_{0.025}(16)$$

$$t = 2.120$$

$$\frac{4x}{\sqrt{1-x^2}} = 2.120$$

Square root on both sides

$$16x^2 \geq (2.120)^2 (1-x^2)$$

$$16x^2 \geq 4.4944(1-x^2)$$

$$16x^2 \geq 4.4944 - 4.4944x^2$$

$$16x^2 + 4.4944x^2 \geq 4.4944$$

$$20.4944x^2 \geq 4.4944$$

$$x^2 \geq \frac{4.4944}{20.4944}$$

$$20.4944$$

$$\sqrt{x^2} \geq \sqrt{0.22}$$

$$x \geq 0.4690$$

LONG

- ① Candidates for scientific post gave particulars of their university degrees and their hobbies. The degrees were in either university degrees and their hobbies could be classified roughly as music, Craftwork, reading or drama. The data are presented concisely in the following Contingency Table

Categories	Maths.	Chem	Phy
Music	24	83	17
Craftwork	11	62	28
Reading	32	121	34
Drama	10	26	44

Discuss the association between the two criteria of classification the degree and hobbies

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Solution:-**Hypotheses:-**

H_0 : The two criteria of classification are independent:

H_1 : The two criteria of classification are not independent.

Level of Significance:-**Test-Statistics:-**

$$\chi^2 = \sum \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Calculation:-

Categories	Maths	Chem	phy	Total
Music	24	83	17	124
Craftswork	11	62	28	101
Reading	32	121	34	187
Drama	10	26	44	80
Total	77	292	123	492

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observed frequency o_i	Expected frequency e_i	$o_i - e_i$	$(o_i - e_i)^2$	$(o_i - e_i)^2 / e_i$
24	$\frac{77 \times 124}{492} = 19.4$	4.6	21.16	1.09
11	$\frac{77 \times 101}{492} = 15.8$	-4.8	23.04	1.46
32	$\frac{77 \times 187}{492} = 29.3$	2.7	7.29	0.25
10	$\frac{77 \times 80}{492} = 12.5$	-2.5	6.25	0.5
83	$\frac{292 \times 124}{492} = 73.6$ 59.8	9.4	88.36	1.20
62	$\frac{292 \times 101}{492} = 59.9$	2.1	4.41	0.07
121	$\frac{292 \times 187}{492} = 111$	10	100	0.90
26	$\frac{292 \times 80}{492} = 47.5$	-21.5	462.25	9.73
17	$\frac{123 \times 124}{492} = 31$	-14	196	6.32
28	$\frac{123 \times 101}{492} = 25.25$	2.7	7.29	0.29
34	$\frac{123 \times 187}{492} = 46.7$	-12.7	161.29	3.45
44	$\frac{123 \times 80}{492} = 20$	24	576	28.8

Total

$$\chi^2 = 54.06$$

Critical Region:-

$$\chi^2 \geq \chi^2_{0.05} (r-1)(c-1)$$

$$\chi^2 \geq \chi^2_{0.05} (4-1)(3-1)$$

$$\chi^2 \geq \chi^2_{0.05} (3)(2)$$

$$\chi^2 \geq \chi^2_{0.05} (6)$$

$$\chi^2 \geq 12.59$$

$$54.06 \geq 12.59$$

Conclusion :-

The value of χ^2 is greater than the critical region, so our null hypothesis (H_0) is reject.

- ② perform the ANOVA on the data and analyse the treatment means using LSD test with a $\alpha = 0.05$

	Treatment					
Block	1	2	3	4	5	6
1	1	3	6	4	3	2
2	1	4	4	8	5	1
3	3	6	7	8	4	3
4	2	3	2	3	2	1

Solution:-

Hypothesis:-

$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5 = \mu_6$$

H_a all the six means are

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equal.

H_1 : All the six means are not equal.

Significance level:-

$$\alpha = 0.05$$

Test-statistics:-

$$F_1 = \frac{S_1^2}{S_2^2}$$

Calculation:-

Treatments

Block	1	2	3	4	5	6	$\bar{T}_i$	$\bar{T}_i^2$	$\sum X_{ij}^2$
1	1(1)	3(9)	6(36)	4(16)	8(64)	2(4)	19	361	75
2	1(1)	4(16)	4(16)	8(64)	5(25)	1	23	529	123
3	3(9)	6(36)	7(49)	8(64)	4(16)	3(9)	31	961	183
4	2(4)	3(9)	2(4)	3(9)	2(4)	1	13	169	31
$\bar{T}_j$	7	16	19	23	14	7	86	2020	412
$\bar{T}_j^2$	49	256	361	529	196	49	1440		↓
$\sum X_{ij}$	15	70	105	153	54	15	402	→	checked

$$\text{Total ss} = \sum X_{ij}^2 - \frac{\bar{T}_{..}^2}{rc}$$

$$= 412 - \frac{(86)^2}{(4 \times 6)}$$

$$= 412 - \frac{7396}{24}$$

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$$= 412 - 308.167$$

$$\boxed{\text{Total SS} = 103.833}$$

$$\text{Between SS} = \frac{T_j^2}{r} = 308.167$$

RSS

$$= \frac{1440}{4} - 308.167$$

4

$$= 360 - 308.167$$

$$\boxed{\text{Between SS} = 51.833}$$

$$\text{SSC} = \frac{T_c^2}{r} = 167$$

$$= \frac{2020}{6} - 308.167$$

6

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$$= 336.67 - 308.167$$

$$\boxed{\text{SSC} = 28.499}$$

$$\text{Error SS} = \text{TSS} - (\text{BSS} + \text{CSS})$$

$$= 103.833 - (51.833 + 28.499)$$

$$= 103.833 - 80.332$$

$$\text{Error SS} = 23.501$$

The calculation are presented
in the following ANOVA
Table.

Sources of Variation	d.f	SS	MS	F
Between Treatments	$(t-1)$ 5	51.833	10.37	$F_1 = 10.37$ $\frac{10.37}{1.57}$ $F_1 = 6.61$
Between Blocks	$(R-1)$ 3	28.499	9.496	---
Error	$(R-1)(t-1)$ 15	23.501	1.57	---
Total	23	103.833	---	---

Critical region: $F \geq F_{0.05}(5, 15)$

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 $F_1 \geq F_{0.05}(5, 15)$

$F_1 \geq 2.90$

$6.61 \geq 2.90$

Conclusion:-

The calculated value F_1 is greater than the critical region. So, the null hypothesis (H_0) is rejected. Accept the alternative hypothesis (H_1).

③ Find the linear regression equation from the following data:

$$\Sigma x = 725, \Sigma y = 1011, \Sigma x^2 = 44475$$

$$\Sigma y^2 = 85905, \Sigma xy = 61685$$

Assuming normality test the hypothesis:

$$H_0: \alpha = 32 \text{ vs } H_1: \alpha \neq 32$$

$$H_0: \beta = 0 \text{ vs } H_1: \beta \neq 0$$

Solution:-

Hypothesis:-

$$H_0: \alpha = 32$$

$$H_1: \alpha \neq 32$$

Level of Significance:-

$$\alpha = 0.05$$

Test-Statistic:-

$$t = \frac{a - \alpha}{S_a}$$

$$\therefore S_a = S_{y \cdot x} \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{\Sigma (x - \bar{x})^2}}$$

Calculation:-

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Linear regression equation

$$\hat{Y} = a + bx$$

$$b = \frac{n \sum XY - \sum X (\sum Y)}{n \sum X^2 - (\sum X)^2}$$

put the values in formula

$$b = \frac{(12)(61685) - (725)(1011)}{(12)(44475) - (725)^2}$$

$$b = \frac{740220 - 732975}{533700 - 525625}$$

$$b = \frac{7245}{8075}$$

$$b = 0.89$$

$$\bar{Y} = \frac{\sum Y}{n} = \frac{1011}{12}$$

$$\bar{Y} = 84.25$$

$$\bar{X} = \frac{\sum X}{n} = \frac{725}{12} = 60.42$$

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$$84.25 = a + 0.89(60.42)$$

$$84.25 = a + 53.7708$$

$$84.25 - 53.7708 = a$$

$$\boxed{a = 30.48}$$

$$S_a = S_y \cdot n \sqrt{\frac{1}{n} - \frac{(\bar{x})^2}{\sum (x - \bar{x})^2}}$$

$$S_y \cdot n = \sqrt{\frac{\sum (y - \bar{y})^2}{n - 2}}$$

$$= \sqrt{\frac{\sum y^2 - a \sum y - b \sum xy}{n - 2}}$$

$$= \sqrt{\frac{(85905) - (30.48)(1011) - (0.89)(61685)}{12 - 2}}$$

$$= \sqrt{\frac{85905 - 30815.28 - 54,899.65}{10}}$$

$$= \sqrt{\frac{190.07}{10}}$$

$$= \sqrt{19.007}$$

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$$[Sx.x = 4.359]$$

$$S_a = (4.359) \left[\frac{1}{n} + \frac{(60.42)^2}{(725 - 60.41)^2} \right]$$

$$= (4.359) \left[\frac{1}{12} + \frac{3649.36}{664.59} \right]$$

$$= (4.359) [0.0833 + 5.49115]$$

$$= (4.359) (5.5744)$$

$$= (4.359) (2.36)$$

$$[S_a = 10.2872]$$

$$t = \frac{a - d}{S_a}$$

$$= \frac{30.48 - 32}{10.2872}$$

$$= \frac{-1.52}{10.2875}$$

$$[t = -0.15]$$

Critical region:-

$$|t| \geq t_{0.05}(n-2)$$

$$|t| \geq t_{0.05}(12-2)$$

$$|t| \geq t_{0.05,10}$$

$$|t| \geq \cancel{3.169} \quad 1.812$$

$$|t| \geq 1.812$$

$$0.15 \geq 1.812$$

Conclusion:-

Since the calculated value of t is less than the critical region and does not fall in critical region. So, null hypothesis is accept

ii- Hypothesis:-

$$H_0: \rho_{URCP} = 0$$

$$H_1: \rho \neq 0$$

Level of significance:-

$$\alpha = 0.05$$

Test-statistic:-

$$t = \frac{b - \beta}{s_b}$$

$$\therefore s_b^2 = \frac{\sum (Y - \hat{Y})^2}{n - 2 \sum (X - \bar{X})^2}$$

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Calculation:-

$$S_b^2 = \frac{\sum (Y - \bar{Y})^2}{n - 2} \frac{\sum (X - \bar{X})^2}{\sum (X - \bar{X})^2}$$

$$= \frac{\sum Y^2 - a \sum Y - b \sum XY}{n - 2} \frac{\sum (X - \bar{X})^2}{\sum (X - \bar{X})^2}$$

$$\sum (X - \bar{X})^2 = \sum X^2 - \frac{(\sum X)^2}{n}$$

$$= 44475 - \frac{(61685)^2}{12}$$

$$= 44475 - 52562.5$$

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$$= 44475 - 43862.68$$

$$\boxed{\sum (X - \bar{X})^2 = 612.91}$$

$$\sum Y^2 - a \sum Y - b \sum XY = 85905 - (30.42 \times 1011) - (0.89)(61685)$$

$$= 85905 - 30815.22$$

$$- 54899.65$$

$$= 190.07$$

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$$S_b^2 = \frac{190.07}{(12-2)(672.91)}$$

$$= \frac{190.07}{(10)(672.91)}$$

$$= \frac{190.07}{6729.1}$$

$$S_b^2 = 0.028224$$

$$\sqrt{S_b^2} = \sqrt{0.028224}$$

$$S_b = 0.168$$

$$t = \frac{b - \beta_{PUACP}}{S_b}$$

$$= \frac{0.89 - 0}{0.168}$$

$$= \frac{0.89}{0.168}$$

$$= 5.5625$$

$$t = 5.5625$$

Critical region:-

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$$|t| \geq t_{0.05}(10)$$

$$|t| \geq 1.812$$

$$5.5625 \geq 1.812$$

Conclusion:-

Since the calculated value of t_c is greater than the critical region. So, the null hypothesis (H_0) is reject and alternative hypothesis (H_1) is accepted.

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